Assignment and Viva on Matrices, Complex Variable and Fourier Analysis

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Assignment: (10 Marks)

1. Find the inverse of the matrix by any method:

$$\begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

2. Solve the following system using Gauss-Elimination or Gauss-Jordan method and explain why are you using this?

$$2x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + z = 0$$

3. Find the eigen-value and eigen vector of the following

$$\begin{bmatrix} -5 & 1 & 2 \\ 0 & -2 & 0 \\ 4 & 2 & -3 \end{bmatrix}$$

4. When a system of linear equation will be consistent, inconsistent and no solution for homogeneous system?

5. Use De Moivre's theorem to prove

$$cos^4\theta = \frac{1}{8}(cos4\theta + 2cos2\theta + 3)$$

6. If $e^{i\theta} = cos\theta + isin\theta$, then prove that

$$sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

7. Solve the equation

$$z^4 = 1 + i\sqrt{3}$$

8. Consider the function : $f(x) = x + x^2; \quad -\pi \leq x \leq \pi$

   1. Compute the Fourier coefficients a_0, a_n and b_n in detail, showing all steps for each integral.
   2. Using the coefficients found, write the complete Fourier series expansion of $f(x)$.

Viva (10 Marks)

- Brief explanation of the theory and every method you used to solve your assignment.
- Questions may include basic concepts or any topic related to this course.